

# Numerical Analysis I

## Interpolation Methods — Central Differences & Divided Differences

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### Stirling's Interpolation Formula

#### Central Differences — Overview

In central difference interpolation, we use differences from *both sides* of the interpolating point  $x_0$ . The parameter  $p = \frac{x - x_0}{h}$  lies in the range  $[-\frac{1}{2}, \frac{1}{2}]$ . Stirling's formula is most accurate when  $p$  is close to 0.

#### Stirling's Difference Table

The difference table is centred at  $x_0$  with entries at  $x_{-3}, x_{-2}, x_{-1}, x_0, x_1, x_2, x_3$ :

| $x$      | $y$      | $\Delta y$      | $\Delta^2 y$      | $\Delta^3 y$      | $\Delta^4 y$      | $\Delta^5 y$      |
|----------|----------|-----------------|-------------------|-------------------|-------------------|-------------------|
| $x_{-2}$ | $y_{-2}$ |                 |                   |                   |                   |                   |
|          |          | $\Delta y_{-2}$ |                   |                   |                   |                   |
| $x_{-1}$ | $y_{-1}$ |                 | $\Delta^2 y_{-2}$ |                   |                   |                   |
|          |          | $\Delta y_{-1}$ |                   | $\Delta^3 y_{-2}$ |                   |                   |
| $x_0$    | $y_0$    |                 | $\Delta^2 y_{-1}$ |                   | $\Delta^4 y_{-2}$ |                   |
|          |          | $\Delta y_0$    |                   | $\Delta^3 y_{-1}$ |                   | $\Delta^5 y_{-2}$ |
| $x_1$    | $y_1$    |                 | $\Delta^2 y_0$    |                   | $\Delta^4 y_{-1}$ |                   |
|          |          | $\Delta y_1$    |                   | $\Delta^3 y_0$    |                   |                   |
| $x_2$    | $y_2$    |                 | $\Delta^2 y_1$    |                   |                   |                   |
|          |          | $\Delta y_2$    |                   |                   |                   |                   |
| $x_3$    | $y_3$    |                 |                   |                   |                   |                   |

### Derivation

Stirling's formula is the **arithmetic mean** of Gauss Forward and Gauss Backward formulae.

**Gauss Forward formula:**

$$y_p = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_{-1} + \frac{(p+1)p(p-1)}{3!}\Delta^3 y_{-1} + \frac{(p+1)p(p-1)(p-2)}{4!}\Delta^4 y_{-2} + \dots$$

**Gauss Backward formula:**

$$y_p = y_0 + p\nabla y_0 + \frac{p(p+1)}{2!}\nabla^2 y_0 + \frac{(p+1)p(p-1)}{3!}\nabla^3 y_0 + \frac{(p+2)(p+1)p(p-1)}{4!}\nabla^4 y_0 + \dots$$

Adding both and dividing by 2:

#### Stirling's Interpolation Formula

$$y_p = y_0 + p \left[ \frac{\Delta y_0 + \Delta y_{-1}}{2} \right] + \frac{p^2}{2!} \Delta^2 y_{-1} + \frac{p(p^2-1)}{3!} \left[ \frac{\Delta^3 y_{-1} + \Delta^3 y_{-2}}{2} \right] + \frac{p^2(p^2-1)}{4!} \Delta^4 y_{-2} + \frac{p(p^2-1)(p^2-4)}{5!} \left[ \frac{\Delta^5 y_{-2} + \Delta^5 y_{-3}}{2} \right] + \dots$$

$$\text{where } p = \frac{x - x_0}{h}.$$

**Notes:** (1) Odd-factorial terms use the *mean* of two differences. (2) Even-factorial terms use a single difference. (3) Odd terms start with  $p$ 's; even terms start with  $p^2$ 's.

## Solved Example

## Example — Stirling's Formula

Given:

| $x$       | 1.0    | 1.2    | 1.4    | 1.6    | 1.8    |
|-----------|--------|--------|--------|--------|--------|
| $y = e^x$ | 2.7183 | 3.3201 | 4.0552 | 4.9530 | 6.0496 |

Find  $y(1.3)$  using Stirling's formula. Here  $h = 0.2$ , take  $x_0 = 1.4$ .

Difference Table:

| $x$ | $y$    | $\Delta y$ | $\Delta^2 y$ | $\Delta^3 y$ | $\Delta^4 y$ |
|-----|--------|------------|--------------|--------------|--------------|
| 1.0 | 2.7183 |            |              |              |              |
|     |        | 0.6018     |              |              |              |
| 1.2 | 3.3201 |            | 0.1333       |              |              |
|     |        | 0.7351     |              | 0.0294       |              |
| 1.4 | 4.0552 |            | 0.1627       |              | 0.0066       |
|     |        | 0.8978     |              | 0.0360       |              |
| 1.6 | 4.9530 |            | 0.1987       |              |              |
|     |        | 1.0966     |              |              |              |
| 1.8 | 6.0496 |            |              |              |              |

Applying Stirling's Formula:

$$p = \frac{1.3 - 1.4}{0.2} = -0.5; \quad y_0 = 4.0552, \quad \Delta y_0 = 0.8978, \quad \Delta y_{-1} = 0.7351, \quad \Delta^2 y_{-1} = 0.1627, \\ \Delta^3 y_{-1} = 0.0360, \quad \Delta^3 y_{-2} = 0.0294, \quad \Delta^4 y_{-2} = 0.0066.$$

$$\begin{aligned} y_{1.3} &= 4.0552 + (-0.5) \left[ \frac{0.8978 + 0.7351}{2} \right] + \frac{(-0.5)^2}{2} (0.1627) \\ &\quad + \frac{(-0.5)((-0.5)^2 - 1)}{6} \left[ \frac{0.0360 + 0.0294}{2} \right] + \frac{(-0.5)^2((-0.5)^2 - 1)}{24} (0.0066) \\ &= 4.0552 - 0.5(0.8165) + 0.125(0.1627) \\ &\quad + \frac{(-0.5)(-0.75)}{6} (0.0327) + \frac{(0.25)(-0.75)}{24} (0.0066) \\ &= 4.0552 - 0.4082 + 0.0203 + 0.00204 - 0.0000516 \\ &\approx \mathbf{3.6693} \end{aligned}$$

## Result

$$y(1.3) \approx 3.6693 \quad (\text{Exact: } e^{1.3} = 3.6693) \quad \checkmark$$

## Bessel's Interpolation Formula

### When to Use Bessel's Formula

Bessel's formula is a central difference formula most accurate when  $p \approx \frac{1}{2}$ , i.e., when the interpolation point lies near the *midpoint* between two tabulated values.

### Derivation

Starting from Gauss Forward formula and substituting:

$$\Delta^3 y_{-1} = \Delta^2 y_0 - \Delta^2 y_{-1}, \quad \Delta^5 y_{-2} = \Delta^4 y_{-1} - \Delta^4 y_{-2}$$

### Bessel's Interpolation Formula

$$y_p = \frac{y_0 + y_1}{2} + \left(p - \frac{1}{2}\right) \Delta y_0 + \frac{p(p-1)}{2!} \cdot \frac{\Delta^2 y_{-1} + \Delta^2 y_0}{2} \\ + \frac{\left(p - \frac{1}{2}\right) p(p-1)}{3!} \Delta^3 y_{-1} + \frac{p(p-1)(p+1)(p-2)}{4!} \cdot \frac{\Delta^4 y_{-2} + \Delta^4 y_{-1}}{2} + \dots$$

where  $p = \frac{x - x_0}{h}$  and  $y_1$  is the value at  $x_1 = x_0 + h$ .

At  $p = \frac{1}{2}$ : the terms with  $(p - \frac{1}{2})$  vanish, leaving only even-order difference terms — making the formula particularly efficient at the midpoint.

## Solved Example

## Example — Bessel's Formula

Given:

|     |    |    |    |    |
|-----|----|----|----|----|
| $x$ | 10 | 20 | 30 | 40 |
| $y$ | 46 | 66 | 81 | 93 |

Find  $y(25)$  using Bessel's formula. Take  $x_0 = 20$ ,  $h = 10$ .

Difference Table:

| $x$       | $y$       | $\Delta y$ | $\Delta^2 y$ | $\Delta^3 y$ |
|-----------|-----------|------------|--------------|--------------|
| 10        | 46        |            |              |              |
|           |           | 20         |              |              |
| <b>20</b> | <b>66</b> |            | -5           |              |
|           |           | 15         |              | <b>2</b>     |
| <b>30</b> | <b>81</b> |            | -3           |              |
|           |           | 12         |              |              |
| 40        | 93        |            |              |              |

$$p = \frac{25 - 20}{10} = 0.5; \quad y_0 = 66, \quad y_1 = 81, \quad \Delta y_0 = 15, \quad \Delta^2 y_{-1} = -5, \quad \Delta^2 y_0 = -3, \quad \Delta^3 y_{-1} = 2.$$

$$\begin{aligned}
 y_{25} &= \frac{66 + 81}{2} + (0.5 - 0.5)(15) + \frac{(0.5)(0.5 - 1)}{2} \cdot \frac{-5 + (-3)}{2} + \frac{(0.5 - 0.5)(0.5)(0.5 - 1)}{6}(2) \\
 &= 73.5 + 0 + \frac{(0.5)(-0.5)}{2} \cdot (-4) + 0 \\
 &= 73.5 + (-0.125)(-4) \\
 &= 73.5 + 0.5 = \mathbf{74}
 \end{aligned}$$

## Result

$$y(25) = \mathbf{74}$$

## Lagrange's Interpolation Formula

## Lagrange Polynomials

Lagrange's method constructs an interpolating polynomial through  $n + 1$  points that may be **unequally spaced**. No difference table is required.

## Formula

## General Form

Given  $(x_0, y_0), (x_1, y_1), \dots, (x_n, y_n)$ , the interpolating polynomial is:

$$P(x) = \sum_{i=0}^n L_i(x) y_i = L_0(x) y_0 + L_1(x) y_1 + \dots + L_n(x) y_n$$

where the **basis polynomials** are:

$$L_i(x) = \prod_{\substack{j=0 \\ j \neq i}}^n \frac{x - x_j}{x_i - x_j}$$

Explicitly:

$$L_0(x) = \frac{(x - x_1)(x - x_2) \cdots (x - x_n)}{(x_0 - x_1)(x_0 - x_2) \cdots (x_0 - x_n)}$$

$$L_1(x) = \frac{(x - x_0)(x - x_2) \cdots (x - x_n)}{(x_1 - x_0)(x_1 - x_2) \cdots (x_1 - x_n)}$$

**Key property:**  $L_i(x_j) = \delta_{ij}$  (1 if  $i = j$ , 0 otherwise). Each basis polynomial equals 1 at its own node and 0 at all others.

## Solved Example

## Example — Lagrange's Formula

Given:  $y = x^3$ 

| $x$ | 1 | 2 | 3  | 4  |
|-----|---|---|----|----|
| $y$ | 1 | 8 | 27 | 64 |

Find  $y(1.5)$  using Lagrange's interpolation.Compute basis polynomials at  $x = 1.5$ :

$$L_0(1.5) = \frac{(1.5 - 2)(1.5 - 3)(1.5 - 4)}{(1 - 2)(1 - 3)(1 - 4)} = \frac{(-0.5)(-1.5)(-2.5)}{(-1)(-2)(-3)} = \frac{-1.875}{-6} = 0.3125$$

$$L_1(1.5) = \frac{(1.5 - 1)(1.5 - 3)(1.5 - 4)}{(2 - 1)(2 - 3)(2 - 4)} = \frac{(0.5)(-1.5)(-2.5)}{(1)(-1)(-2)} = \frac{1.875}{2} = 0.9375$$

$$L_2(1.5) = \frac{(1.5 - 1)(1.5 - 2)(1.5 - 4)}{(3 - 1)(3 - 2)(3 - 4)} = \frac{(0.5)(-0.5)(-2.5)}{(2)(1)(-1)} = \frac{0.625}{-2} = -0.3125$$

$$L_3(1.5) = \frac{(1.5 - 1)(1.5 - 2)(1.5 - 3)}{(4 - 1)(4 - 2)(4 - 3)} = \frac{(0.5)(-0.5)(-1.5)}{(3)(2)(1)} = \frac{0.375}{6} = 0.0625$$

Interpolated value:

$$\begin{aligned} P(1.5) &= 1(0.3125) + 8(0.9375) + 27(-0.3125) + 64(0.0625) \\ &= 0.3125 + 7.5 - 8.4375 + 4 = \mathbf{3.375} \end{aligned}$$

## Result

$$y(1.5) = \mathbf{3.375} \quad (\text{Exact: } (1.5)^3 = 3.375) \quad \checkmark$$

## Advantages and Disadvantages

| Advantages                      | Disadvantages                                  |
|---------------------------------|------------------------------------------------|
| Works for unequally spaced data | Adding a new point requires full recomputation |
| No difference table required    | Computationally expensive for large $n$        |
| Direct polynomial form          | Susceptible to Runge's phenomenon              |

## Newton's Divided Difference Method

**Motivation**

Lagrange's formula has the drawback that if a new interpolation point is inserted, all coefficients must be recomputed. Newton's Divided Difference method overcomes this by building the polynomial *incrementally*.

**Divided Differences**

Let  $y_0, y_1, \dots, y_n$  be values of  $y = f(x)$  at  $x_0, x_1, \dots, x_n$  (not necessarily equally spaced).

**First divided differences:**

$$f[x_i, x_{i+1}] = \frac{y_{i+1} - y_i}{x_{i+1} - x_i}$$

**Second divided differences:**

$$f[x_i, x_{i+1}, x_{i+2}] = \frac{f[x_{i+1}, x_{i+2}] - f[x_i, x_{i+1}]}{x_{i+2} - x_i}$$

**$k$ -th divided difference:**

$$f[x_0, x_1, \dots, x_k] = \frac{f[x_1, x_2, \dots, x_k] - f[x_0, x_1, \dots, x_{k-1}]}{x_k - x_0}$$

**Newton's Divided Difference Formula**

$$P(x) = f[x_0] + (x - x_0) f[x_0, x_1] + (x - x_0)(x - x_1) f[x_0, x_1, x_2] + \dots + (x - x_0)(x - x_1) \dots (x - x_{n-1}) f[x_0, \dots, x_n]$$



## Solved Example

## Example — Newton's Divided Difference

Given:

| $x$    | 0 | 1 | 3  | 4   |
|--------|---|---|----|-----|
| $f(x)$ | 5 | 6 | 50 | 105 |

Find  $f(2)$  using Newton's divided difference method.

Divided Difference Table:

| $x$ | $f[x]$ | $f[\cdot, \cdot]$         | $f[\cdot, \cdot, \cdot]$ | $f[\cdot, \cdot, \cdot, \cdot]$ |
|-----|--------|---------------------------|--------------------------|---------------------------------|
| 0   | 5      |                           |                          |                                 |
|     |        | $\frac{6-5}{1-0} = 1$     |                          |                                 |
| 1   | 6      |                           | $\frac{22-1}{3-0} = 7$   |                                 |
|     |        | $\frac{50-6}{3-1} = 22$   |                          | $\frac{11-7}{4-0} = 1$          |
| 3   | 50     |                           | $\frac{55-22}{4-1} = 11$ |                                 |
|     |        | $\frac{105-50}{4-3} = 55$ |                          |                                 |
| 4   | 105    |                           |                          |                                 |

Newton's polynomial:

$$\begin{aligned}
 P(x) &= 5 + (x-0)(1) + (x-0)(x-1)(7) + (x-0)(x-1)(x-3)(1) \\
 &= 5 + x + 7x(x-1) + x(x-1)(x-3) \\
 &= 5 + x + 7x^2 - 7x + x^3 - 4x^2 + 3x \\
 &= x^3 + 3x^2 - 3x + 5
 \end{aligned}$$

Find  $f(2)$ :

$$\begin{aligned}
 P(2) &= 5 + (2)(1) + (2)(1)(7) + (2)(1)(-1)(1) \\
 &= 5 + 2 + 14 - 2 = \mathbf{19}
 \end{aligned}$$

Verify:  $P(2) = 8 + 12 - 6 + 5 = 19$  ✓Spot checks:  $P(0) = 5$ ,  $P(1) = 6$ ,  $P(3) = 50$ ,  $P(4) = 105$  all match. ✓

## Result

$$f(2) = \mathbf{19}$$

### Comparison of Interpolation Methods

| Method              | Spacing | Best for                    | Key Feature                    |
|---------------------|---------|-----------------------------|--------------------------------|
| Stirling's          | Equal   | $p \approx 0$ (middle)      | Mean of Gauss fwd & bwd        |
| Bessel's            | Equal   | $p \approx 0.5$ (mid-point) | Best at midpoint               |
| Lagrange's          | Unequal | Any point                   | No difference table needed     |
| Newton's Div. Diff. | Unequal | Any point                   | Incremental, efficient updates |

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